

Magnetostatics

Page No: _____

Date: _____

Magnetic lines are always closed in nature.

→ Biot-Savart law:

also called as Ampere's law for current element. It gives diff. magnetic field intensity.

$\vec{dH}$ due to diff. current element $I \cdot d\vec{l}$, consider a filament through which current of I amp. is passing. We are interested in finding field intensity H at pt. P .

→ Statement:

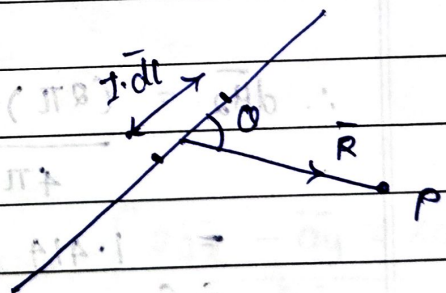
Magnitude of dH at pt. P is prop. to product of current, diff. length $d\vec{l}$, the sine of the angle betn. the filament & line connecting diff. length to the point of interest P .

It is inv. prop. to the sq. of distance from the filament to point P .

$$dH = \frac{(I \cdot d\vec{l}) \sin \theta}{4\pi R^2}$$

$$\vec{dH} = \frac{(I \cdot d\vec{l}) \times \vec{a}_R}{4\pi R^2} \quad (\text{A/m})$$

$$\therefore H = \oint \frac{I \cdot d\vec{l} \times \vec{a}_R}{4\pi R^2} \rightarrow \underline{\text{line}}$$

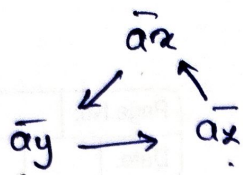


$$\vec{dH} = \frac{\vec{J} \cdot dV \times \vec{a}_R}{4\pi R^2}$$

→ volume

$$\vec{dH} = \frac{\vec{K} \cdot dS \times \vec{a}_R}{4\pi R^2}$$

→ Surface



$$\begin{aligned} \vec{a}_x \times \vec{a}_x &= \vec{a}_y & (\text{in dir. of arrow}) \\ \vec{a}_x \times \vec{a}_y &= -\vec{a}_z & (\text{opp. of arrow}) \end{aligned}$$

Page No.	
Date.	

Q] Estimate the incremental field $d\vec{H}_2$ at pt. 'P₂' caused by a source at 'P₁' of $I d\vec{l}$

- i) $2\pi \vec{a}_y$ nA-m, P₁(4, 0, 0) & P₂(0, 3, 0)
- ii) $2\pi \vec{a}_x$ nA-m, P₁(4, -2, 3) & P₂(0, 3, 0)

→ Soln.:

i) $\hat{a}_R = \frac{-4\vec{a}_x + 3\vec{a}_y}{\sqrt{4^2 + 3^2}} = \frac{-4\vec{a}_x + 3\vec{a}_y}{5}$

$$\therefore d\vec{H}_2 = \frac{I d\vec{l} \times \hat{a}_R}{4\pi R^2}$$

$$= \frac{(2\pi)(10^{-6}) \vec{a}_y}{4\pi(5)^2} \times \left[\frac{-4\vec{a}_x + 3\vec{a}_y}{5} \right]$$

$$= [-12\vec{a}_x - 16\vec{a}_y] \text{ (nA/m)}$$

ii) $\hat{a}_R = \frac{-4\vec{a}_x + 5\vec{a}_y - 3\vec{a}_z}{\sqrt{50}}$

$$\therefore d\vec{H}_2 = \frac{(2\pi)(10^{-6}) \vec{a}_x}{4\pi(50)} \times \left[\frac{-4\vec{a}_x + 5\vec{a}_y - 3\vec{a}_z}{\sqrt{50}} \right]$$

$$= 1.414 (10^{-9}) \vec{a}_x \times [-4\vec{a}_x + 5\vec{a}_y - 3\vec{a}_z]$$

$$= [-5.656 \vec{a}_y - 7.07 \vec{a}_z] \text{ (nA/m)}$$

Q] Given points $A(1, 2, 4)$, $B(-2, -1, 3)$ & $C(3, 1, -2)$
 let a dipole current element $I = 6A$ and $|dl| = 10^{-4}m$
 be located at A , the dirn. of dl is from A to B .
 Find dH at C .

→ Solution:

$$\bullet \quad I \cdot dl = (6)(10^{-4}) (\bar{a}_{AB})$$

$$\bar{a}_{AB} = \frac{-3\bar{a}_x - 3\bar{a}_y - \bar{a}_z}{\sqrt{19}}$$

$$\begin{aligned} \therefore I \cdot dl \cdot \bar{a}_{AB} &= (6)(10^{-4}) \left[\frac{-3\bar{a}_x - 3\bar{a}_y - \bar{a}_z}{\sqrt{19}} \right] \\ &= -0.413 \bar{a}_x - 0.413 \bar{a}_y - 0.188 \bar{a}_z \quad (mA \cdot m) \end{aligned}$$

• Now, By Biot Savart's law

$$dH = \frac{I \cdot dl \times \bar{a}_{AC}}{4\pi (AC)^2}$$

Now,

$$\bar{a}_{AC} = \frac{2\bar{a}_x - \bar{a}_y - 6\bar{a}_z}{\sqrt{41}}$$

$$\begin{aligned} \therefore dH &= \frac{(-0.413 \bar{a}_x - 0.413 \bar{a}_y - 0.188 \bar{a}_z) \times \left[\frac{2\bar{a}_x - \bar{a}_y - 6\bar{a}_z}{\sqrt{41}} \right]}{4\pi (41)} \\ &= 0.413 \bar{a}_z \end{aligned}$$

Q] using Biot-Savart law, find the vector mag field intensity in cartesian co-ordinates at $P_2 (1.5, 2, 3)$ caused by a current filament of 24 A in the $\bar{a}_x$ dirn. on the x -axis and extending from : @ $x=0$ to $x=6$

(b) $x=6$ to $x=\infty$

(c) $x=-\infty$ to $x=\infty$

→ Solution:

$$\frac{\bar{r}}{r^3} = \frac{\bar{r}_1 - \bar{r}_2}{r^3} = \bar{r}_1 - \bar{r}_2$$

$$\left[\frac{\bar{r}_1 - \bar{r}_2}{r^3} \right] (\bar{a}_x) (dl) = \bar{a}_x \cdot \bar{r}_1 - \bar{r}_2$$

$$(1 \cdot 10^{-7}) \left[\frac{\bar{r}_1 - \bar{r}_2}{r^3} \right] (\bar{a}_x) (dl) = \bar{a}_x \cdot \bar{r}_1 - \bar{r}_2$$

$$\frac{\bar{r}_1 \times \bar{r}_2}{r^3} = \bar{r}_1$$

$$\frac{\bar{r}_1 - \bar{r}_2}{r^3} = \bar{r}_1$$

$$\frac{\bar{r}_1 - \bar{r}_2}{r^3} \left[\frac{\bar{r}_1 - \bar{r}_2}{r^3} \right] (\bar{a}_x) (dl) = \bar{r}_1$$

$$\bar{r}_1 - \bar{r}_2 = \bar{r}_1$$

- * Magnetic Flux (Ψ_m):
 It is found to be The magnetic flux crossing any surface is found by mul. magnetic flux density with area of the surface.

$$\Psi_m = \int_S \vec{B} \cdot d\vec{s}$$

q] A radial field : $\vec{H} = \frac{(2.39)(10^6)}{r} \cos \phi \vec{a}_r$ (A/m)

exists in free space. Find magnetic flux crossing the surface defined by $0 \leq \phi \leq \pi/4$, $0 \leq x \leq 1$ m

→ Solution:

Now,

$$\vec{B} = \mu \vec{H} = \mu_0 \mu_r \vec{H}$$

$$= (4\pi \times 10^{-7}) (1) \left[\frac{2.39 \times 10^6}{r} \right] \cos \phi \vec{a}_r$$

$$= 30 (10^{-1}) \frac{\cos \phi}{r} \vec{a}_r = \frac{3 \cos \phi}{r} \vec{a}_r$$

also, $d\vec{s} = r \cdot d\phi \cdot dx \vec{a}_r$

$$\therefore \Psi_m = \int_S \vec{B} \cdot d\vec{s}$$

$$= \int_{\phi=0}^{\pi/4} \int_{x=0}^1 \frac{3 \cos \phi}{r} \vec{a}_r \cdot r d\phi dx \vec{a}_r$$

$$= 3 \int_0^{\pi/4} \cos \phi \cdot d\phi \int_0^1 dx$$

$$= 3 [+\sin \phi]_0^{\pi/4} [x]_0^1$$

$$\therefore \Psi_m = 2.12 \text{ Wb}$$

Q) $\vec{B} = 2.5 \left[\sin \frac{\pi x}{2} \right] e^{-2y} \cdot \vec{a}_x$; find total Ψ_m
 $y \geq 0, 0 \leq x \leq 2$

→ Solution:

$$\Psi_m = \iint \vec{B} \cdot d\vec{s}$$

$$\because x=0, \therefore d\vec{s} = dx dy \vec{a}_x$$

$$\therefore \Psi_m = \int_0^2 \int_0^\infty 2.5 \left[\sin \frac{\pi x}{2} \right] e^{-2y} \cdot dx dy$$

$$= 2.5 \left[-\cos \frac{\pi x}{2} \right]_0^2 \left[\frac{e^{-2y}}{-2} \right]_0^\infty$$

$$= \frac{2.5}{\pi} [\cos \pi - \cos 0] [-1 - 0]$$

$$= \frac{2.5}{\pi/2} [-1] [-1] = \frac{5}{\pi} \text{ wb}$$

* Ampere's Law:

$$\oint \vec{H} \cdot d\vec{l} = I$$

ie. line integral of $\vec{H}$ around a single closed path is equal to I enclosed by that path.

→ Cross products:

$$\bullet \nabla \times \vec{H} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix}$$

Cartesian

$$\bullet \nabla \times \vec{H} = \begin{vmatrix} \frac{1}{r} \vec{a}_r & \vec{a}_\phi & \frac{1}{r} \vec{a}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ H_r & r H_\phi & H_z \end{vmatrix}$$

Cylindrical

$$\bullet \nabla \times \vec{H} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \vec{a}_r & r \vec{a}_\theta & r \sin \theta \vec{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ H_r & r H_\theta & r \sin \theta H_\phi \end{vmatrix}$$

Spherical

$$\nabla \times \vec{H} = \vec{J}$$

q] $\vec{H}$ due to a current source is given by:

$$\vec{H} = y \cos(2x) \vec{a}_x + (y + e^x) \vec{a}_y, \text{ describe the } \vec{J} \text{ at origin.}$$

→ Solution:

$$\vec{J} = \nabla \times \vec{H} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y \cos 2x & 0 & y + e^x \end{vmatrix}$$

$$= \vec{a}_x [1 - 0] - \vec{a}_y [e^x - 0] + \vec{a}_z [0 - \cos 2x]$$

$$= \vec{a}_x - e^x \vec{a}_y - \cos 2x \vec{a}_z$$

∴ at origin $(0, 0, 0) \equiv (x, y, z)$.

$$\vec{J} = \vec{a}_x - \vec{a}_y - \vec{a}_z$$

q] $\vec{H} = yz(x^2 + y^2) \vec{a}_x - y^2 xz \vec{a}_y + 4x^2 yz \vec{a}_z$ (A/m)

Determine: i) $\vec{J}$ at $(5, 2, -3)$

ii) Show that $\nabla \cdot \vec{B} = 0$

→ Soln.:

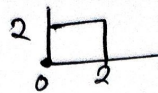
a) $\nabla \times \vec{H} = \vec{J}$

$$\therefore \vec{J} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x^2 + y^2)yz & -y^2 xz & 4x^2 yz \end{vmatrix}$$

$$= \vec{a}_x [8x^2 y + y^2 x] - \vec{a}_y [8xy^2 - (x^2 + y^2)y] + \vec{a}_z [-y^2 x - x^2 x - 2y^2 x]$$

∴ $\vec{J}$ at $(5, 2, -3)$:

$$\therefore \vec{J} = \frac{8(5)^2(2) + (2)^2(-3)}{420} \vec{a}_x - \frac{8(5)(2)^2 - (5^2 + 2^2)(2)}{420} \vec{a}_y + \frac{123}{420} \vec{a}_z$$



$$b) \quad \vec{B} = \mu \vec{H} = \mu_0 \vec{H}$$

$$= (4\pi)(10^{-7}) \vec{H}$$

Then prove $\nabla \cdot \vec{B} = 0$

* Stokes's Theorem:

States that integration of any vector around a closed path is always equal to the integration of curl of that vector throughout the surface enc. by that path.

$$\oint \vec{H} \cdot d\vec{l} = \int_S (\nabla \times \vec{H}) \cdot d\vec{s}$$

Q] verify Stoke's Thm. for the field $\vec{H} = 6xy \vec{a}_x - 3y^2 \vec{a}_y$ and the rectangular path around the region lies betn $2 \leq x \leq 5$, $-1 \leq y \leq 1$, $z=0$.
let the +ve dirn. of $d\vec{s}$ be $\vec{a}_z$

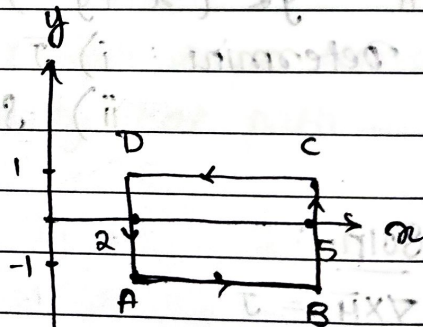
Solution:

RHS • Now,

$$\nabla \times \vec{H} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 6xy & -3y^2 & 0 \end{vmatrix}$$

$$= \vec{a}_x(0) - \vec{a}_y(0-0) + \vec{a}_z(0-6x)$$

$$= -6x \cdot \vec{a}_z$$



$$\therefore \int_S (\nabla \times \vec{H}) \cdot d\vec{s} = - \int_{y=-1}^1 \int_{x=2}^5 6x \cdot d\vec{s}$$

$$= -6 [y]_{-1}^1 \left[\frac{x^2}{2} \right]_2^5 = (-6)(+2) \left[\frac{25}{2} - \frac{4}{2} \right]$$

$$= -6 [2] = -12$$

$$\begin{aligned}
 \oint \vec{H} \cdot d\vec{l} &= \int_{AB} \vec{H} \cdot d\vec{l} + \int_{BC} \vec{H} \cdot d\vec{l} + \int_{CD} \vec{H} \cdot d\vec{l} + \int_{DA} \vec{H} \cdot d\vec{l} \\
 &= \int_2^5 (6xy - 3y^2) dx + \int_1^2 (-3y^2) dy + \int_5^2 (6xy) dx \\
 &\quad + \int_1^{-1} (-3y^2) dy \\
 &= \left[\frac{6x^2y}{2} - \frac{3y^2x}{2} \right]_2^5 + \left[-\frac{3y^3}{3} \right]_1^2 + \left[\frac{6x^2y}{2} \right]_5^2 + \left[-\frac{3y^3}{3} \right]_1^{-1} \\
 &= 3[25-4] - [1+1] + 3[4-25] - [-1-1] \\
 &= -3(21) - (2) + 3(-21) - (-2) \\
 &= -126
 \end{aligned}$$

$$\therefore \text{RHS} = \text{LHS}$$

$$\therefore \oint \vec{H} \cdot d\vec{l} = \int_V (\nabla \times \vec{H}) d\vec{s}$$

Q] Evaluate the closed line integral of $\vec{H}$ from $P_1(5,4,1)$ to $P_2(5,6,1)$ to $P_3(0,6,1)$ to $P_4(0,4,1)$ and back to P_1 using straight line segments, if $\vec{H} = 0.1y^3 \vec{a}_x + 0.4x \vec{a}_y$ (A/m), determine $(\nabla \times \vec{H})_z$ at the center.

→ Solution:

$$(\nabla \times \vec{H})_z = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 0.1y^3 & 0 & 0.4x \end{vmatrix}$$

$$= a_x(0) - a_y(0.4 - 0) + a_z(-0.3y^2)$$

$$= -0.4\vec{a}_y - 0.3y^2 \vec{a}_z$$

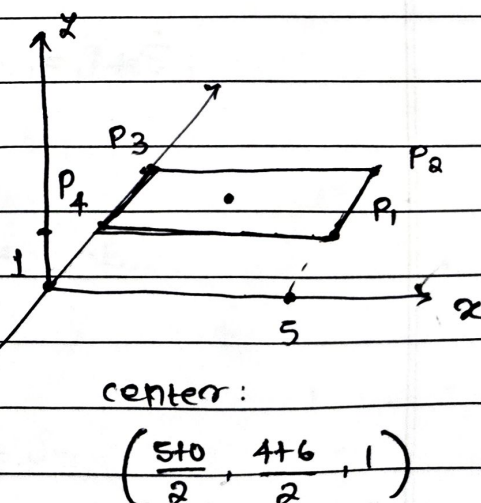
(only $\vec{a}_z$ term)

$\therefore$ line integral of $\vec{H}$

$\therefore \nabla \times \vec{H}$ at center:

$$= -0.4\vec{a}_y - 0.3(5)^2 \vec{a}_z$$

$$= -4.5 \vec{a}_z$$



center:

$$\left(\frac{5+0}{2}, \frac{4+6}{2}, 1 \right)$$

$$= (2.5, 5, 1)$$

* properties of curl of vector: $\nabla \cdot (\nabla \times \vec{A}) = 0$

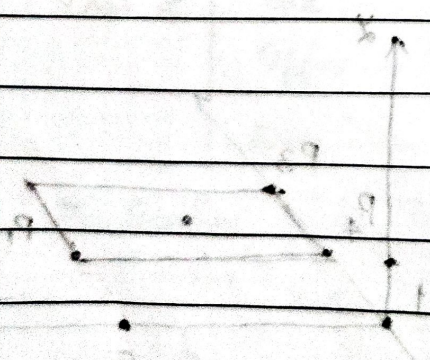
• $\nabla \cdot (\nabla \times \vec{A}) = 0$, $\text{div}(\text{curl}) = 0$

• $\nabla \times (\nabla \phi) = 0$, $\text{curl}(\text{grad}) = 0$

• If, $\nabla \times (\vec{A} + \vec{B}) = \nabla \times \vec{A} + \nabla \times \vec{B}$

$\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$
(ie $\text{grad}(\text{div}) - \dots$)

• $\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B})$



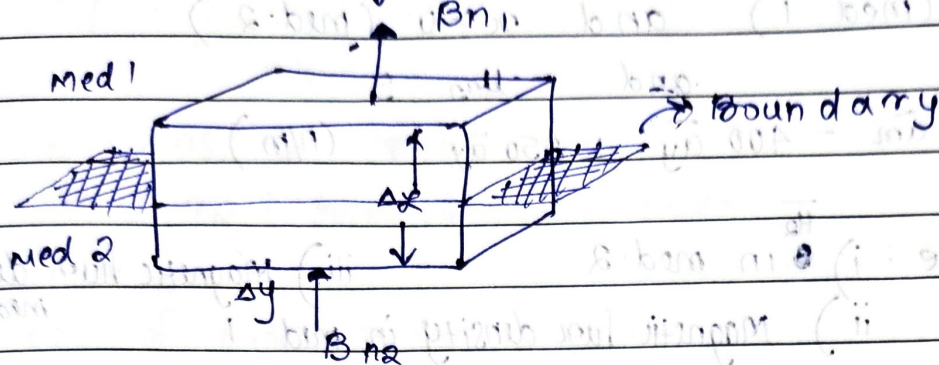
$\vec{A}$	$\vec{B}$	$\vec{C}$
$0\hat{i} + 0\hat{j} + 0\hat{k}$	$0\hat{i} + 0\hat{j} + 0\hat{k}$	$0\hat{i} + 0\hat{j} + 0\hat{k}$
$0\hat{i} + 0\hat{j} + 0\hat{k}$	$0\hat{i} + 0\hat{j} + 0\hat{k}$	$0\hat{i} + 0\hat{j} + 0\hat{k}$
$0\hat{i} + 0\hat{j} + 0\hat{k}$	$0\hat{i} + 0\hat{j} + 0\hat{k}$	$0\hat{i} + 0\hat{j} + 0\hat{k}$

$(0\hat{i} + 0\hat{j} + 0\hat{k}) \cdot (0\hat{i} + 0\hat{j} + 0\hat{k}) - (0\hat{i} + 0\hat{j} + 0\hat{k}) \cdot (0\hat{i} + 0\hat{j} + 0\hat{k}) = 0$

$\vec{A} \cdot \vec{B} = 0$

$(\text{only } \vec{A} \text{ term})$

* Magnetic Boundary conditions



→ BC for normal comp. of $\vec{B}$:

$$\oint \vec{B} \cdot d\vec{s} = 0$$

• NOW,

$$\text{LHS} = \oint \vec{B} \cdot d\vec{s}$$

$$= \int_{\text{Top}} \vec{B} \cdot d\vec{s} + \int_{\text{Bot}} \vec{B} \cdot d\vec{s} + \int_{\text{Left}} \vec{B} \cdot d\vec{s} + \int_{\text{Right}} \vec{B} \cdot d\vec{s} + \int_{\text{Front}} \vec{B} \cdot d\vec{s} + \int_{\text{Back}} \vec{B} \cdot d\vec{s}$$

• NOW, at end ---

$$B_{n1} = B_{n2}$$

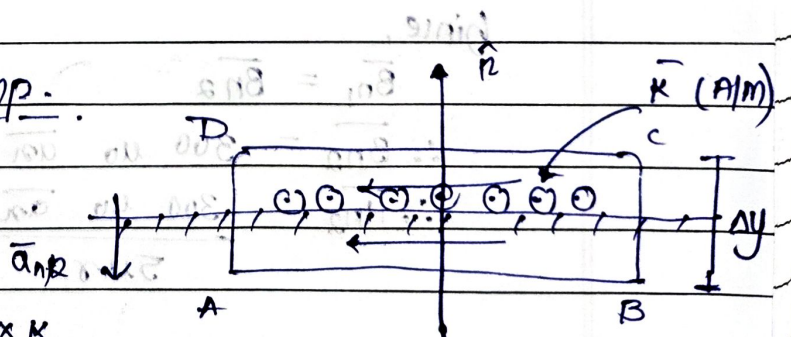
($\therefore$ continuous across interface)

→ BC for tangential comp.

$$H_{t1} - H_{t2} = K$$

$\therefore$ In vector form:

$$\vec{H}_{t1} - \vec{H}_{t2} = \vec{a}_{n12} \times K$$



• If no current along interface

$$H_{t1} - H_{t2} = 0$$

$$H_{t1} = H_{t2}$$

Q] 2 homogeneous, have interface at $x=0$,
 $x < 0$ (med. 1) and $x > 0$ (med. 2)

$$\mu_1 = 2 \quad \text{and} \quad \mu_2 = 5$$

$$\vec{H}_1 = 150 \vec{a}_x - 400 \vec{a}_y + 250 \vec{a}_z \quad (\text{A/m})$$

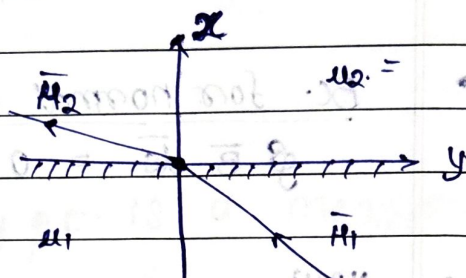
Determine: i) $\vec{H}_2$ in med. 2 iii) Magnetic flux density in med. 2
 ii) Magnetic flux density in med. 1

→ Solution:

∵ interface at $x=0$

∵ Surface is yz plane

and Normal dirn. is along x axis.



$$\text{Now, } \vec{H}_{t1} = -400 \vec{a}_y + 250 \vec{a}_z$$

$$\vec{H}_{n1} = 150 \vec{a}_x$$

$$\text{Now, } \vec{H}_{t1} = \vec{H}_{t2}$$

$$\therefore \vec{H}_{t2} = -400 \vec{a}_y + 250 \vec{a}_z$$

also,

$$\vec{B}_{n1} = \mu_1 \vec{H}_{n1}$$

$$= \mu_0 (2) (150 \vec{a}_x) = 300 \mu_0 \vec{a}_x \quad (\text{wb/m}^2)$$

hence,

$$\vec{B}_{n1} = \vec{B}_{n2}$$

$$\therefore \vec{B}_{n2} = 300 \mu_0 \vec{a}_x \quad (\text{wb/m}^2)$$

$$\therefore \vec{H}_{n2} = \frac{300 \mu_0 \vec{a}_x}{5 \mu_0} = 60 \vec{a}_x \quad (\text{A/m})$$

$$\vec{H}_2 = \vec{H}_{n2} + \vec{H}_{t2} = 60 \vec{a}_x - 400 \vec{a}_y + 250 \vec{a}_z$$

$$\therefore \vec{B}_1 = \mu_1 \vec{H}_1 = 2 \mu_0 (150 \vec{a}_x - 400 \vec{a}_y + 250 \vec{a}_z)$$

$$\therefore \vec{B}_2 = \mu_2 \vec{H}_2 = 5 \mu_0 (60 \vec{a}_x - 400 \vec{a}_y + 250 \vec{a}_z)$$

* List of formula on inductance:

1)
$$L = \frac{\mu_0 \mu_r (N^2 h)}{2\pi} \ln \left[\frac{r_2}{r_1} \right] \quad (\mu)$$

N : turns, h : height

r_1 & r_2 : inner & outer radii

2) Area of cross section: S

$$L = \frac{\mu_0 \mu_r N^2}{2\pi a} (S) \quad (\mu)$$

3)
$$L = \frac{\mu_0 \mu_r N^2}{d} (S) \quad (\mu/m)$$

d : length

4)
$$L = \frac{\mu_0}{8\pi} + \frac{\mu_0}{2\pi} \ln \left[\frac{b}{a} \right] \quad (\mu/m)$$

now, $(\mu_0/8\pi)$: internal L / wt. length

a, b : in. out. radii

5) 2 || wires with separation d & radius of each wire: a

$$L = \frac{\mu_0}{4\pi} + \frac{\mu_0}{\pi} \ln \left[\frac{d}{a} \right] \quad (\mu/m)$$

9) $\epsilon_r = 18$, $\delta = 0$
 $\mu_r = 80$, $a = 1 \text{ mm}$, $b = 4 \text{ mm}$

→ Solution:

$$L = \frac{\mu_0 \mu_r}{2\pi} \ln \left(\frac{b}{a} \right) \quad (L \text{ of coaxial cable})$$

$$= 22.1807 \mu(\mu/m)$$

$$\therefore L = 22.1807 (10) = 221.807 \mu H/m$$